# AI based Optimization of Global Maximum Power Point Tracking for Photovoltaic Systems during Partial Shading Conditions

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2024 Advances in Science and Engineering Technology International Conferences (ASET) | 979-8-3503-4413-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/ASET60340.2024.10708722

***Abstract*— This paper aims to investigate the suitability of Artificial Intelligence (AI) based algorithms for optimizing the Global Maximum Power Point Tracking (GMPPT) performance in Photovoltaic (PV) systems during partial shading conditions (PSC). The performance of AI based techniques such as Genetic Algorithm (GA), Fuzzy Logic Control (FLC), Partial swarm optimization (PSO), Artificial Neural Network (ANN) and Adaptive Neuro Fuzzy Inference Systems (ANFIS) will be examined in this paper. A range of PV system configurations, such as 3 panel ,4 panel and 6 panel PV strings and various DC-DC converter topologies, including buck and boost converters, are utilized to test the scalability and variability of the designs. For evaluating the effectiveness of GMMP tracking during PSC a PV system is modelled and simulated using MATLAB SIMULINK. Fuzzy Logic Control and Artificial Neural Network, Adaptive Neuro Fuzzy Inference Systems (ANFIS) based MPPT are implemented using Fuzzy Toolbox, Neural Network Toolbox and ANFIS toolbox in MATLAB. The outcome of the study shows that the GA algorithm exhibits instability and oscillations during partial shading conditions (PSC), failing to track the Global MPP (GMPP) under PSC reliably. The FLC algorithm struggles to track the GMPP during PSC accurately. On the other side, PSO demonstrates a good tracking performance, achieving a GMPP tracking efficiency of 90.23% on average, though it does not track under certain PSCs the average MPPT tracking efficiency of ANN is 77.71% for the six cases. However, ANN is unable to track GMPP and is unstable during PSC. Out of six partial shading tests conducted, ANFIS MPPT was able to track the GMPP in three specific PSC scenarios.**

***Keywords—Photovoltaic, Maximum Power Point Tracking, Genetic Algorithm, Fuzzy Logic Control, Partial swam optimization, Artificial Neural Networks, Adaptive Neuro Fuzzy Inference Systems.***

## I. INTRODUCTION

The advancement of Artificial Intelligence based control in renewable energies could help solve the ongoing energy crisis. Aside from others, solar energy is becoming the point of attraction for a new era, as evidenced by the renewables worldwide status report issued in 2019 [1]. Maximal Power Point Tracking (MPPT) converters are used to obtain the maximum energy from a PV array. Over the years, a variety of MPPT techniques have been introduced [2],[3],[4] which intends to track the Maximum Power Point (MPP), the optimal operating point of DC-DC converter for maximum power transfer. However, under PSC, in the presence of multiple peaks conventional techniques struggled to efficiently track the Global MPP [5]. Due to the inherent learning capacity, Intelligent MPPT algorithms can precisely differentiate Global MPP and Local MPP by ensuring the impedance between the PV array and the load is optimally matched [6]. This study conducts comparative analysis of AI based MPPT techniques which employs soft computing techniques like the Particle Swarm Optimization (PSO), Genetic Algorithm (GA), Fuzzy Logic Control (FLC) and Artificial Neural Networks (ANN) [9] [10],[11].

## II. GMPP TRACKING DURING PSC

Hotspot issues arise when partial shading effects cause the polarity of the voltage across shaded cells to shift from positive to negative. Bypass diodes can be used to eliminate these issues, however doing so causes numerous spikes in the power voltage characteristics, which is another issue. Tracking the MPP point and enabling the PV to operate at maximum power are the goals here. When there are several local peaks in the power voltage curve of the PV system, the real problem is to identify the global peak and continue following the GMPP.

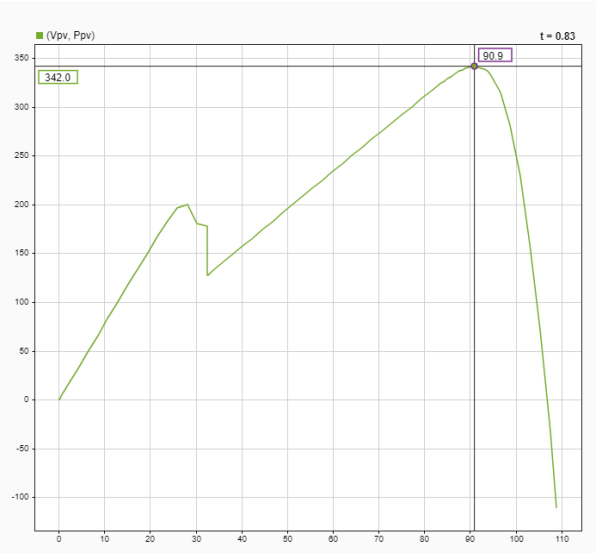



Fig. 1. PV Curve with multiple peaks during PSC

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For a PV string with four panels, when two panels are shadowed with a 500W/m$^2$ as seen in Fig. 1., there will be two peaks in the PV curve. AI based algorithms are utilized in this study to track GMPP under partial shading effects. A PV system modelled using MATLAB Simulink is employed in this study for the analysis. To measure the Algorithm's performance, the model is tested under various irradiance conditions The results thus acquired are evaluated to measure the efficiency of the MPPT controller. Fig. 2 depicts the proposed model block diagram. It basically consist of an MPPT controller, PV array, DC-DC converters and resistive load.

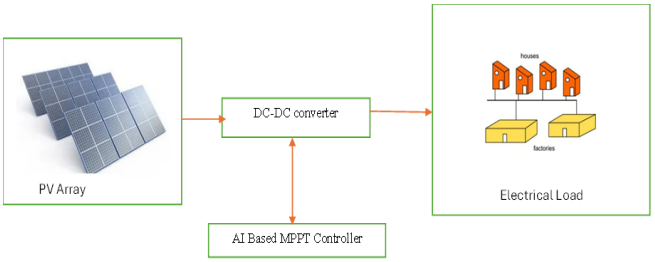



Fig. 2 Proposed model.

## III. AI BASED MPPT TECHNIQUES

To attain the maximum translation efficacy, the photovoltaic panel will operate at the upper peak of the power graph, known as the MPP. Numerous MPPT strategies have also been developed over the last few decades employing both internal and exterior techniques. The most sensible way to employ MPPT algorithms to boost PV panel efficiency is to track the MPP value using an algorithm. In this study, the MPPT method's AI based algorithm is utilized to modify the duty ratio based on instantaneous measurements of the current and voltage in the photovoltaic energy system. The system then regulates it at the DC-DC converter circuit's output. Table I. lists the partial shading employed in this study in W/m$^2$.

TABLE I. PARTIAL SHADING CASES

| PSC | Panel 1 | Panel 2 | Panel 3 | Panel 4 | Panel 5 |
|---|---|---|---|---|---|
| 1 | 1000 | 1000 | 190 | 350 to 190 | 1000 |
| 2 | 500 | 1000 | 340 | 470 to 340 | 1000 |
| 3 | 500 | 1000 | 520 | 560 to 520 | 1000 |
| 4 | | 400 | 670 | 670 | 1000 |
| 5 | | | 780 | 780 | 1000 |
| 6 | | | 890 | 890 | 1000 |

### A. *Genetic Algorithm*

Genetic algorithm (GA) is an AI technique inspired by the method of natural evolution in which a fitness function will reflect the fitness of each member of population. In this case the fitness function is the PV power (Ppv) and the following GA parameters are used. Crossover probability is 0.9, Number of population is 4, $\Delta V = 0.4$ V and $\Delta P = 0.15$. Fig. 3 shows the PV system model used which utilizes a PV string and buck converter. The PV array waveforms are shown in Fig.4.The results in Table II show that the average MPPT tracking efficiency is 72.13% for the six cases. Vmpp is the voltage of the module at the MPP and Pmpp is the power at the MPP.

TABLE II. GA BASED PERFORMANCE ANALYSIS

| PSC | (Vmpp, Pmpp) | V(V) | P(W) | Tracking efficiency | GMP PT |
|---|---|---|---|---|---|
| 1 | 91.7,342.1 | 95.3 | 322.4 | 94.2 | No |
| 2 | 84.8,631.5 | 92.1 | 602.1 | 95.3 | No |
| 3 | 124.1,495.1 | 106.7 | 467.4 | 34.9 | No |
| 4 | 157,567.1 | 160.3 | 169.3 | 29.8 | No |
| 5 | 91.7,342.1 | 104.2 | 310.5 | 90.76 | No |
| 6 | 84.8, 631.5 | 134.6 | 554.5 | 87.81 | No |

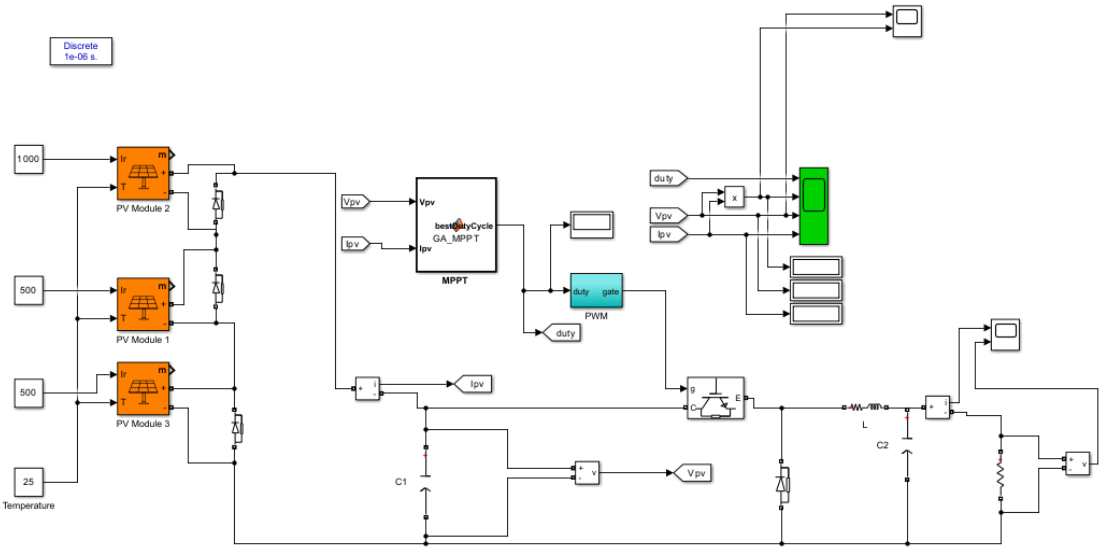



Fig. 3. GA MPPT based PV System

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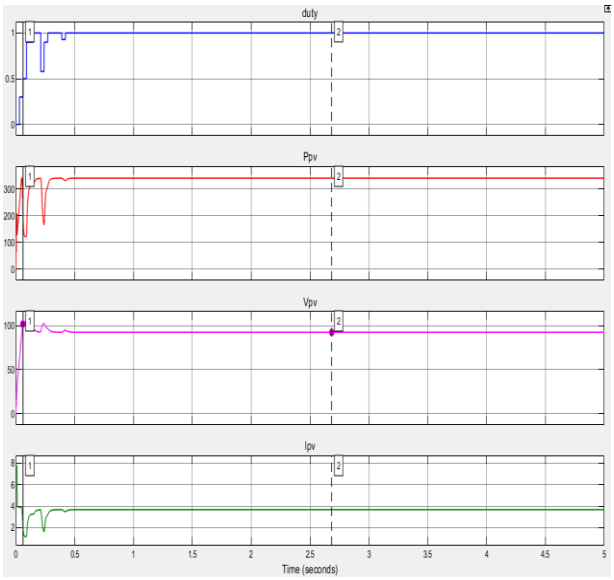



Fig. 4. The waveforms for GA MPPT

The rise time for GA algorithm is better than the conventional method which is between 0.3 and 0.4 seconds. The algorithm is unable to track GMPP and unstable during PSC with oscillations, the GA algorithm is unable to track the GMPP accurately during partial shading conditions. Fig.4. displays the Duty Cycle, PV Array Power, voltage, and current graphs using GA algorithm.

### *B. Fuzzy Logic Controller*

Fuzzy logic control (FLC) principles uses linguistic variables and fuzzy rules to make decisions. FLC can be used to determine the optimal operating point of the solar panel. FLC can handle uncertainties and hence this method making it suitable for tracking the MPP during partial shading scenarios where irradiations weather conditions dynamically changing. The results in Table III show that the MPP tracking efficiency of FLC for the six cases. The rise time for FLC algorithm is between 0.1 to 0.25 seconds. The algorithm is unable to track GMPP and unstable during PSC with oscillations, the FLC algorithm failed to track the Global maxima accurately during PSC under all six PSC cases tested.

TABLE III. FLC BASED PERFORMANCE ANALYSIS

| PSC | (Vmpp, Pmpp) | V(V) | P(W) | Tracking efficiency | GMPPT |
|---|---|---|---|---|---|
| 1 | 91.7,342.1 | 52.2 | 338.5 | 98.94 | No |
| 2 | 84.8, 631.5 | 80.9 | 425.6 | 67.39 | No |
| 3 | 124.1,495.1 | 102.3 | 291.9 | 58.96 | No |
| 4 | 157,567.1 | 114.4 | 289.3 | 51.01 | No |
| 5 | 172.7,1265.9 | 204.5 | 817 | 64.53 | No |
| 6 | 345, 2531 | 235.8 | 1134.5 | 44.82 | No |

### *C. Partial Swam Optimization (PSO)*

The Particle Swarm Optimization (PSO) algorithm is a computational method that simulates the social behavior of birds flocking or fish schooling to find optimal solutions to complex problems. The Simulink model of PSO based PV system with Boost converter is shown in Fig. 5. PSO is robust in many scenarios and highly adaptable to nonlinear optimization problems such as in PV systems. It can find an acceptable solution quickly and with fewer evaluations of the objective function, making it efficient in terms of computational resources. Fig. 6 shows the adjustment of the voltage and current of the $P_{PV}$ to converge to the optimal operating point for case 1. The waveforms in Fig.6 show that the PSO MPPT can track the GMPP in PSC conditions with high tracking efficiency and fast tracking speed. However transient oscillations are high. The results in Table IV show that the average MPPT tracking efficiency of PSO is 90.23% for the six cases. The rise time for PSO algorithm is between 0.4 to 0.6 seconds. The algorithm is unable to track GMPP and unstable during PSC with oscillations.

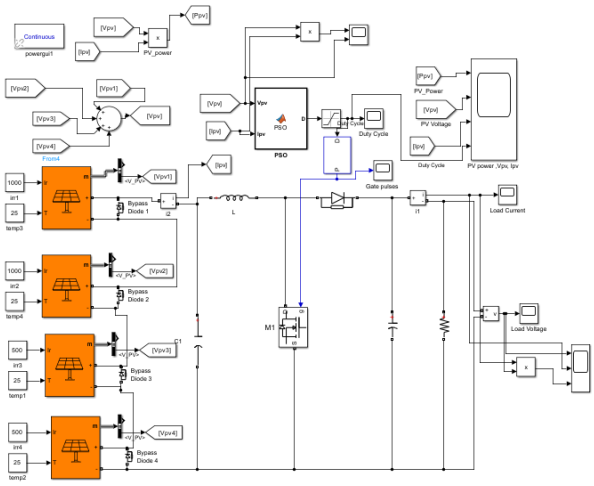



Fig. 5. PSO MPPT based PV System

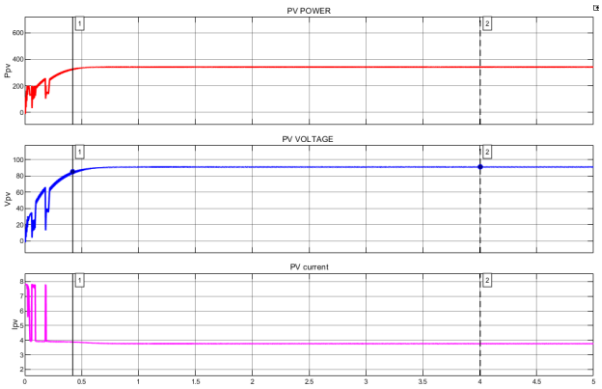



Fig 6. PV array power, voltage and current waveforms

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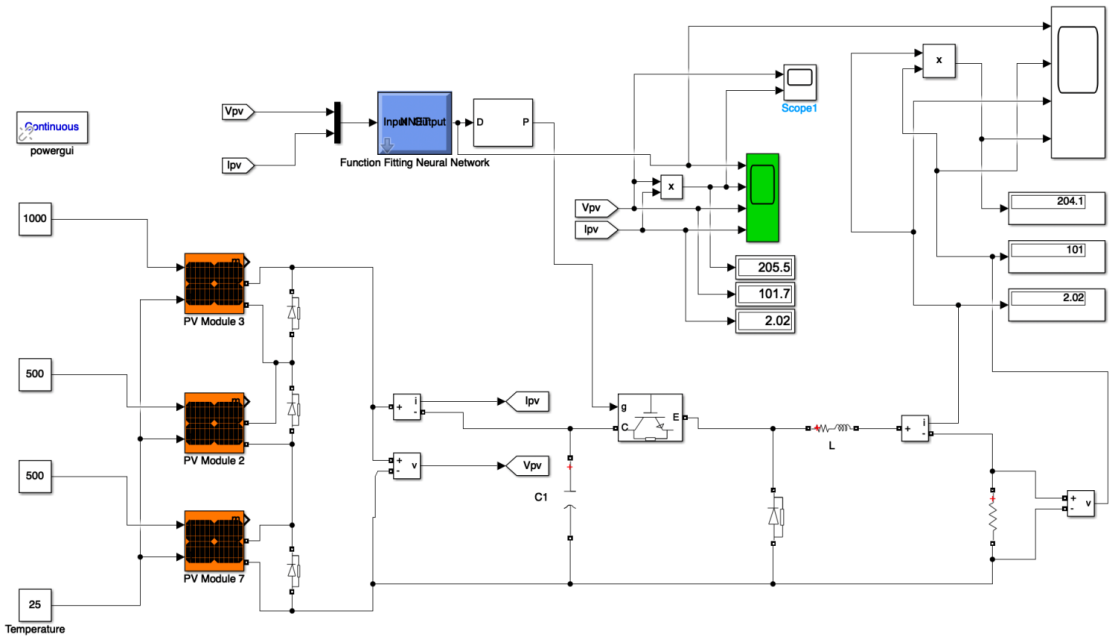



Fig. 7. ANN based PV System

TABLE IV. PSO BASED MPPT PERFORMANCE ANALYSIS

| PSC | (Vmpp, Pmpp) | V(V) | P(W) | Tracking efficiency | GMPPT |
|---|---|---|---|---|---|
| 1 | 91.7,342.1 | 90.94 | 341.4 | 99.79 | Yes |
| 2 | 84.8, 631.5 | 128.3 | 517.83 | 82 | No |
| 3 | 157,567.1 | 105.9 | 467.4 | 82.4 | No |
| 4 | 157,567.1 | 105.6 | 462.8 | 81.6 | No |
| 5 | 172.7, 1265.9 | 174.5 | 1264 | 99.85 | Yes |

### *D. Aritificical Neural Network (ANN)*

Artificial Neural Network inspired by brain neurons trained by supervisory learning algorithm can be employed for predicting the nonlinear behavior of PV during varying irradiation conditions. In the forward pass the initial weights of neural networks are assumed, and the input is passed through the network and the sigmoidal activation function to get the output. Once the output is received it is compared with the original output in the database and the error is obtained. In back propagation algorithm used for neural network training Mean square error is used as performance parameter. In some cases, classification cross entropy is used. Using the chain rule the gradient of the loss function for each weight is calculated. The Stochastic gradient descent algorithm is used to minimize the loss. These steps are repeated for 1000 iterations to minimize the error. The updated weights are now used to predict the output for a new set of data. Fig. 7 shows the MATLAB model and figure 8 shows the error curve and can be seen that the error is around 0.2 which indicates that the prediction percentage will be very high. Fig. 9 and 10 shows the output for the two inputs in two dimensions and three dimensions. For the experiment to evaluate the effectiveness of ANN in tracking GMPP during PSC, a PV system with buck converter is selected as the testbed to evaluate the GMPP tracking efficiency of ANN MPPT algorithm. For demonstrating the tracking of MPPT during PSC the PV characteristics are modelled and simulated using Neural Network toolbox, and MATLAB Simulink as shown in Fig. 7. While ANNs offer a data driven approach to MPPT, their real world performance is influenced by various factors related to their design, training, and external conditions. The ANN MPP controller Neural networks are trained on specific datasets and attempt to generalize their performance to unseen data. For the experiment to evaluate the effectiveness of ANN in tracking GMPP during PSC, a PV system consisting of three panels connected in series was selected as the testbed to evaluate the GMPP tracking efficiency of ANN MPPT algorithm.

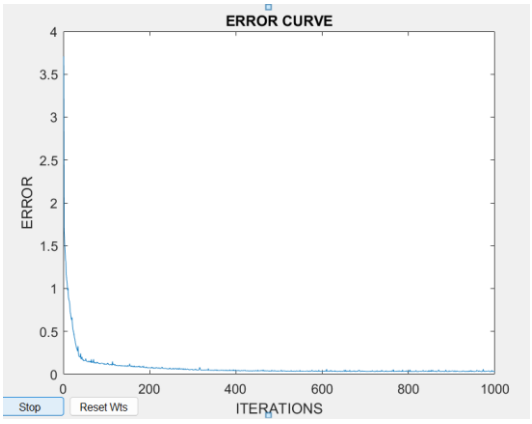



Fig. 8: Error Curve

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While ANNs offer a data driven approach to MPPT, their real world performance is influenced by various factors related to their design, training, and external conditions.

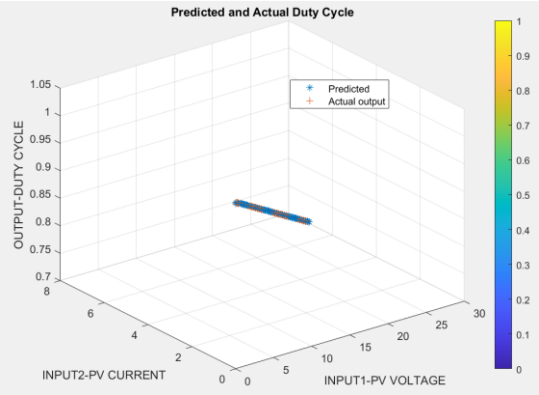



Fig. 9: Predicted and actual Duty Cycle

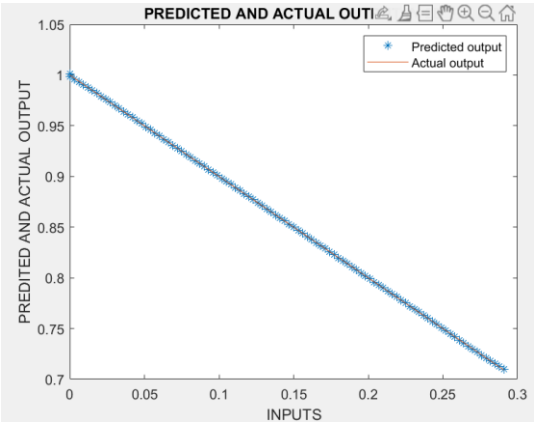



Fig. 10: Actual and predicted output in 2D

The ANN MPP controller Neural networks are trained on specific datasets and attempt to generalize their performance to unseen data. To evaluate the effectiveness of ANN in tracking GMPP during PSC, a PV system consisting of three panels connected in series was selected as the testbed to evaluate the GMPP tracking efficiency of ANN MPPT algorithm. For demonstrating the tracking of MPPT during PSC the PV characteristics are modelled and simulated using Neural Network toolbox, and MATLAB Simulink as. While ANNs offer a data driven approach to MPPT, their real world performance is influenced by various factors related to their design, training, and external conditions. The ANN MPP controller Neural networks are trained on specific datasets and attempt to generalize their performance to unseen data. Fig. 11 shows the simulation results for partial shading testing for Case 1.

TABLE V. ANN BASED MPPT PERFORMANCE ANALYSIS

| PSC | (Vmpp, Pmpp) | V(V) | P(W) | Tracking efficiency | GMPPT |
|---|---|---|---|---|---|
| 1 | 91.12,342.14 | 101.7 | 205.5 | 60.06 | No |
| 2 | 84.8, 631.5 | 134.7 | 360.6 | 82 | No |
| 3 | 124.18,494.96 | 134.6 | 360.4 | 72.81 | No |
| 4 | 156.8,566.86 | 162.5 to 134.6 | 525.4 to 360.4 | 92.68 | Yes |
| 5 | 172.7, 1265.9 | 200.8 | 802.4 | 63.33 | No |
| 6 | 345, 2531 | 369 | 2414 | 95.37 | Yes |

### *E. ANFIS based MPPT controller*

Adaptive Neuro Fuzzy Inference system (ANFIS) is a soft computational tool which is a hybrid of neural networks and fuzzy inference system. ANFIS employs the learning capability of neural networks to optimize the fuzzy inference system (FIS).This results in a system that can accurately model complex nonlinearities as in PV systems. On other hand Fuzzy logic allows ANFIS to handle uncertainty and imprecision inherent in real world applications more effectively than ANN. By optimizing its parameters through learning, ANFIS can achieve higher accuracy and efficiency in modeling and tracking GMPP during PSC compared to ANN or FLC alone. Hybrid models like ANFIS can help in enhancing the GMPP tracking performance. Fig. 12 shows the simulation results for partial shading testing for Case 1. Fig. 13 illustrate an ANFIS based PV System modeled in MATLAB, for investigating the GMPP tracking efficiency of ANIFS MPPT controller.

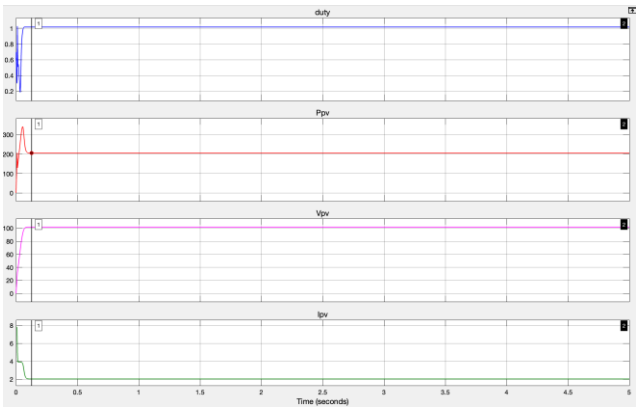



Fig.11. Duty Cycle, PV Power, Voltage and Current ANN Case 1

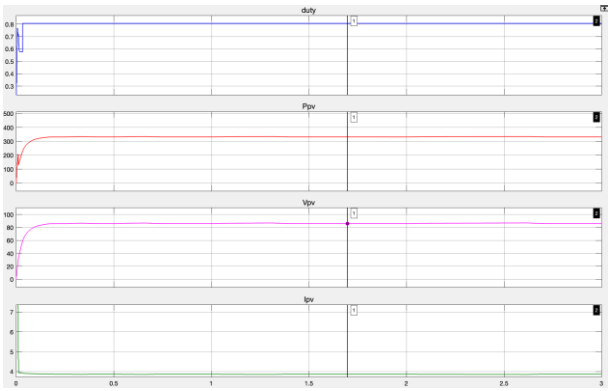



Fig.12. Duty Cycle, PV Power, Voltage and Current for ANFIS Case 1

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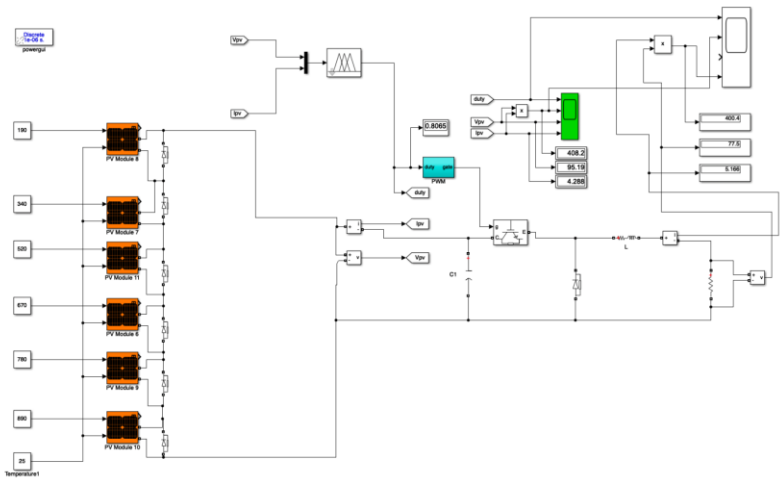


Fig. 13. ANFIS based PV System

TABLE VI. ANFIS BASED MPPT PERFORMANCE ANALYSIS

| PSC | (Vmpp, Pmpp) | V(V) | P(W) | Tracking efficiency | GMPPT |
|---|---|---|---|---|---|
| 1 | 91.7,342.1 | 85.82 | 331.5 | 96.69 | yes |
| 2 | 84.8, 631.5 | 78.27 | 602.1 | 95.34 | yes |
| 3 | 157,567.1 | 95.19 | 408.3 | 71.99 | No |
| 4 | 157,567.1 | 95.19 | 408.2 | 71.98 | No |
| 5 | 172.7, 1265.9 | 200.8 | 802.5 | 63.43 | No |
| 6 | 345, 2531 | 369 | 2414 | 95.7 | Yes |

Table VI shows that out of six partial shading tests conducted, ANFIS MPPT fails to track the GMPP in three specific PSC scenarios. Several factors could influence the tracking performance of ANFIS. During complex shading patterns or rapidly changing shading, the ANFIS will struggle to differentiate between local and global maxima, leading to suboptimal tracking. Though the ANFIS is adaptive and can learn from data, its performance heavily dependents on the quality and range of data it has been trained on. If the training data does not include similar complex PSC conditions, the system might not have learned the appropriate responses.

## IV. RESULTS AND DISCUSSIONS

Table VII illustrates the Qualitative Analysis of MPPT Methods studied in this preliminary research conducted to evaluate the efficiency of existing MPPT techniques. GA uses evolutionary algorithms to find the GMPP but cannot effectively track all PSC scenarios. FLC is introduced which uses a rule based approach to handle uncertainties and nonlinear characteristics of PV systems. PSO is a heuristic approach which can efficiently handle multiple peaks, but it fails to track during some PSC. ANN is capable of learning and adapting to different conditions, making it quite effective in GMPP tracking under varied scenarios. However, the performance depends on the training data quality and network architecture.

TABLE VII. QUALITATIVE ANALYSIS OF AI BASED GMPPT

| Features | FLC | PSO | GA | ANN | ANFIS |
|---|---|---|---|---|---|
| Tracking Accuracy | Medium | High | High | High | High |
| Tracking Speed | Medium | Fast | Medium | Fast | Fast |
| Ability to track MPP | High | High | High | High | High |
| Ability to track GMPP | Poor | Poor | Poor | Poor | Medium |
| Steady state Oscillations | Small | No | No | No | No |

## V. CONCLUSION

In this research the AI based soft computing algorithms GA, FLC, PSO, ANN and ANFIS are investigated for GMPP tracking performance during PSC. Literature review indicated that the traditional methods are simpler and less costly, they are unable to track the GMPP during PSC. Artificial Intelligence based GMPPT techniques were investigated during PSC. However, but failed to track under a few PSC tests conducted. Compared with ANN FLC and PSO, ANFIS is more efficient in tracking GMPP, especially in complex and rapidly changing conditions, but it requires extensive training and is complex to implement. Future research can delve into Reinforcement Learning (RL) algorithm, which can learn directly from interactions with the environment, eliminating the need for dataset and extensive training phases.

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